Factor-base-reduced S3: a checked continuation

Quadratic-twist study, PR2381

25 October 2026; reduced-s3-20261025

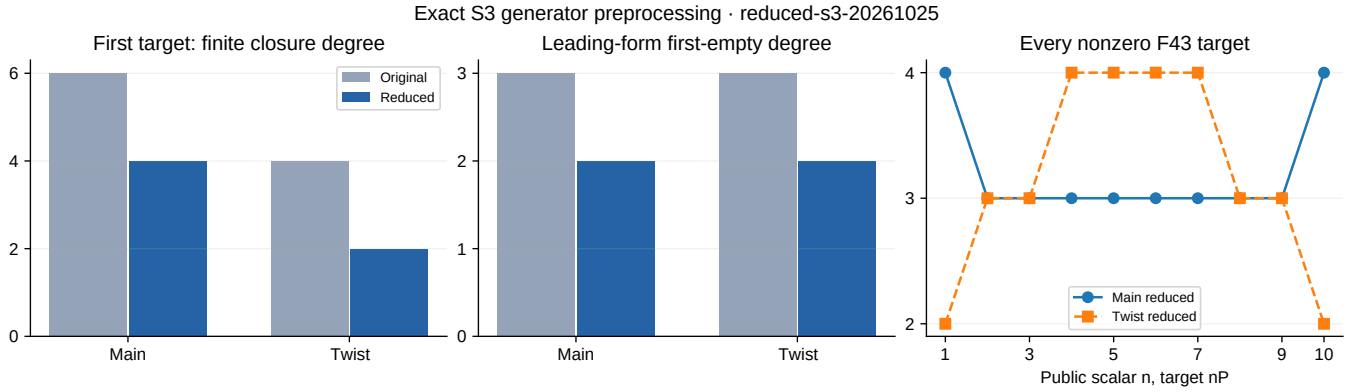
Executed result. C++ and independent Rust replay98 size-two cases, 188 main/twist base comparisons and 1,384 target comparisons. There are 69 joint useful-rank/closure gains and zero joint useful-rank/homogeneous cutoff gains. This is finite exact algebra; no timing or speedup is claimed.

Actual example

Over F_{43} use $E : y^2 = x^3 + x$, its monic twist $T : Y^2 = X^3 + 4X$, subgroup order 11, generators $(31,18)$ and $(24,26)$. The existing models are `icv1-fp6-t0-74779950` and `icv1-fp6-t0-c6a8a47c`. The exact bases are $(4,5),(13,19)$ and $(15,9),(17,6)$.

First target	Main raw	Main reduced	Twist raw	Twist reduced
Independent useful rows	0	0	2	2
Ordered root pairs	0	0	3	3
Finite closure degree	6	4	4	2
Unit-refutation degree	5	3	—	—
Leading-form cutoff	3	2	3	2

The reduced equations are $41xy + 39(x+y) + 27$ and $17xy + 3(x+y) + 41 = 17(x-15)(y-15)$. The twist degree-two matrix has rank 3 in 6 columns and 3 independent root evaluations. The main degree-two matrix has rank 3 with no roots; its first unit occurs at 3, full closure at 4.



Recorded native values, including all ten nonzero targets. The figure source only lays out stored values.

Exact preprocessing and proofs

For $g_x = x^2 - sx + t$ and $g_y = y^2 - sy + t$, write

$$S_3 = (y^2 - 2zy + z^2)g_x + ((s-2z)x + z^2 - t)g_y + h, \quad h = Axy + B(x+y) + C,$$

where $A = s^2 - 4sz - 2z^2 - 2a$, $B = -st + 2zt + sz^2 - 2az - 4b$ and $C = t^2 - 2tz^2 + a^2 - 4bz$. The identity is proved over any commutative ring in Lean, together with equality of the generated ideals and exact equivalence of constrained roots. The fixture's actual leading ideals normalize to $\langle xy, x^2, y^2\rangle$ on both arms, with Hilbert vector $(1,2,0,\ldots)$. Its exact first-empty degree is 2. Because the affine ideal is unchanged, this input-generator cutoff is not intrinsic regularity of the same ideal or its saturated homogenization, nor an asserted F4/F5 solving degree.

Scope, independent checks and transfer

The population consists of all 88 eligible size-two prefixes from the prior 942-model study plus ten F43 subset policies. All 100 cross-pairings of the latter are compared. Every nonzero target, actual signed point lift, coefficient row/rank and homogeneous/affine matrix is independently replayed. Transported and inverse-transported bases retain identical degree/rank profiles. The 69 joint closure observations include repeated base-policy comparisons; they are not 69 independent curves, probabilities or timing trials.

Fifteen new named theorems bring the study to 159 named research theorems and 207 audited declarations across 13 modules. The unmodified scoped gate passes with zero failing/extra trusted axioms; only propext, Classical.choice and Quot.sound occur. New declarations include `ReducedS3.division_identity`, `generated_ideals_equal`, `head_cutoff_exact`, and the explicit F43 remainder and head bindings. The source-bound polynomial certificate `IDC1h77b3276833a18411` passes 32+16 native points with false-identity/source controls. Four rehashed semantic payload controls reject false remainder, degree, rank and omitted case.

The old kernel-checked injective additive map $\phi(O) = O$, $\phi(x,y) = (41x, 11y)$ preserves arbitrary integer scalar relations and sends the main generator to the chosen twist generator. It is not defined by a logarithm lookup. The old 11 independent synthetic recoveries remain historical receipts, not a new recovery run. The generator $(24, 26)$ is explicit and is never relabelled as the older generator's EC1 alias.

Attribution and next experiment

Preprocessing lowers the leading-form cutoff on both fixture arms; it does not establish a twist-only homogeneous advantage in this finite population. Useful yield and finite closure remain target/base dependent. The plot includes losses. The next lead is adaptive reinsertion of newly found lower-degree rows and an explicit solver-degree certificate on the same exact inputs. That follow-up has a draft backlog card and is not executed by this report. All phase and transfer/recovery costs, a target-matched reference and the fastest applicable external implementation remain needed before speed claims.

Reproduction and evidence

From the repository root, run the versioned protocol's native command:
`bash research/quadratic_twist_evidence_20261021/`
`run-reduced-s3.sh FRESH_DIRECTORY`.
Sources, compact canonical records and source pins are in `native/reduced-s3-20261025`; Lean sources are in `lean-reduced`; raw failures, gate receipts, commands and checksums are in `validation/reduced-s3-20261025`. Each native command has a 600-second cap. Python is presentation/metadata only. No prior frozen file or registry, benchmark or leaderboard row is changed. Shared Conductor still exits 127; no reservation or substitute coordinator is claimed. Existing CI/review, integration and mathematical completion gates remain open.